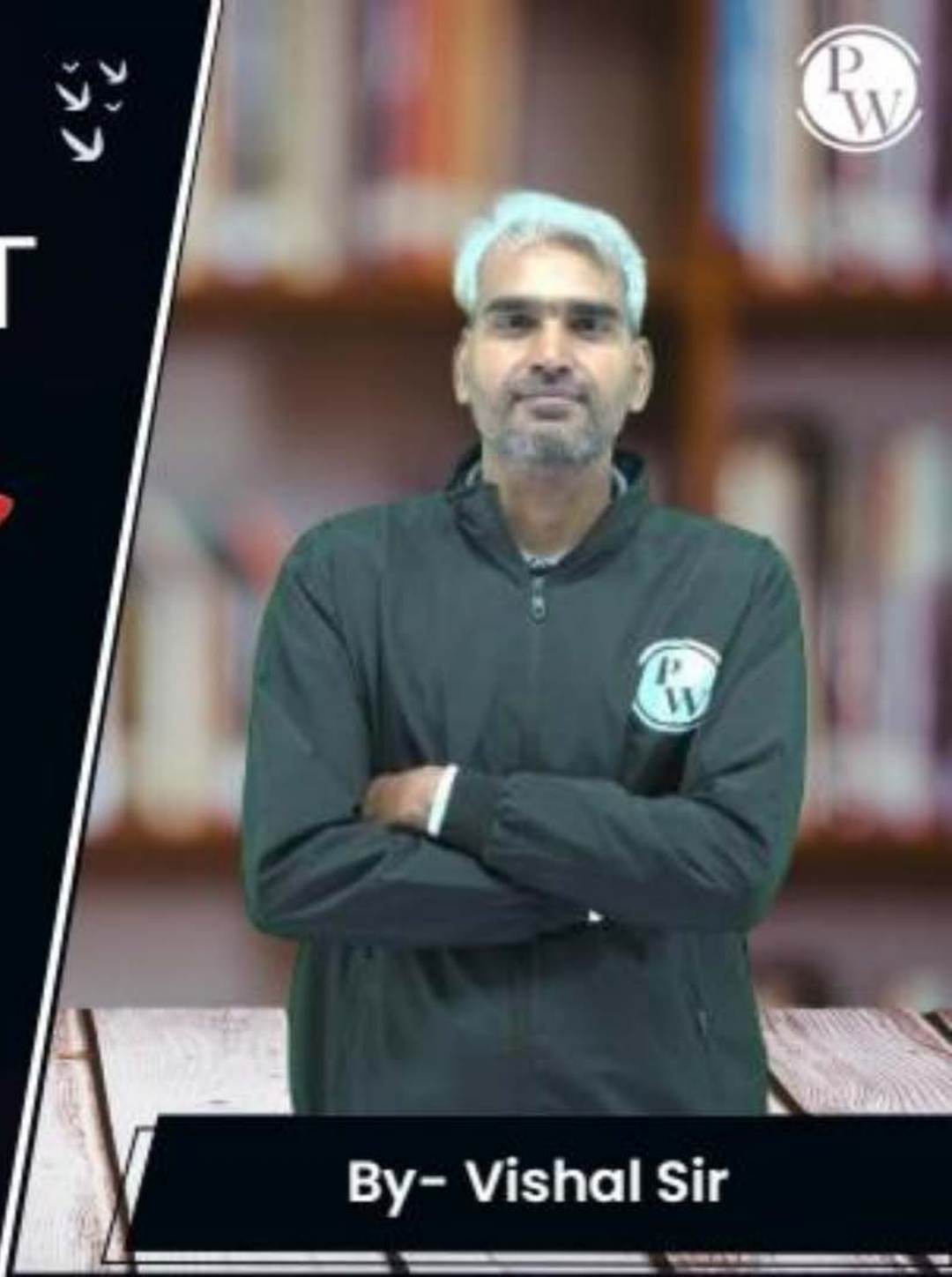
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 08



By- Vishal Sir

Recap of Previous Lecture



Topic Identification of candidate keys in a relation



Slide

Topics to be Covered



- Topic: Membership test
- Topic: Relationship between two FD sets
- Topic: FD set of a sub-relation

Slide



Topic : Membership test

- Membership test is used to check whether a given FD is a member of given FD set or not.

- To check whether $X \rightarrow Y$ is a member of FD set F or not
  (i.e., $F \models X \rightarrow Y$ or not) (or) F can infer $X \rightarrow Y$ or not

We first obtain X^+ (closure of X) w.r.t. FD set F.

If $Y \in X^+$, then $X \rightarrow Y$ is a member of FD set F
otherwise not a member of FD set F

* If $X \rightarrow Y$ is a member of FD set F.
then we can say that
$X \rightarrow Y$ can be inferred from F

#Q. Let FD set F = { A→B, B→C}

Check whether A→C is a member of F or not?

$(A)^+$ w.r.t. F = {A, B, C}

$C \in A^+$ w.r.t. F

∴ A→C is member of F

#Q. Let FD set F = { A→B, B→C}

Check whether B→A is a member of F or not?

$(B)^+$ w.r.t F = {B, C}

$A \notin (B)^+$ w.r.t. F

∴ B→A can not be inferred from F.

#Q. Let FD set F = { AB→C, BC→D}

Check whether AB→D is a member of F or not?

$(AB)^+$ w.r.t. F = {A, B, C, D}

$D \in (AB)^+$ w.r.t F

$\therefore$ AB→D can be inferred from A.

#Q. Let FD set F = { AB→C, C→A}

Check whether C→B is a member of F or not?

$(C)^+$ w.r.t. F = { C, A }

$B \notin (C)^+$ w.r.t F

∴ B→C is not a member of F

Note:-

* Let F is a non-Empty set,

The set of all functional dependencies that can be inferred from F can be denoted by F^+

#Q. Let FD set $F = \{ A \rightarrow B, B \rightarrow C \}$

Find F^+

Possible FDs with three attributes A, B & C are :-

$A \rightarrow B$ ✓ {Member of F}
$A \rightarrow C$ ✓ {— " —}
$B \rightarrow A$ ✗ Not a member of F
$B \rightarrow C$ ✓ {Member of F}
$C \rightarrow A$ ✗
$C \rightarrow B$ ✗
$A \rightarrow BC$ ✓
$B \rightarrow AC$ ✗
$C \rightarrow AB$ ✗

$AB \rightarrow C$ ✓
$AC \rightarrow B$ ✓
$BC \rightarrow A$ ✗

$F^+ = \{ A \rightarrow B, A \rightarrow C, B \rightarrow C, A \rightarrow BC, AB \rightarrow C, AC \rightarrow B \}$



Topic : Relationship between two FD sets

Let F and G are any two FD sets.

- If all the FDs of FD set F are member of FD set G, then $F \subseteq G$
  If G Covers (or) F, then $F \subseteq G$
  If All FDs of F can be inferred (or) from set G, then $F \subseteq G$
- If all the FDs of FD set G are member of FD set F, then $G \subseteq F$
  If F Covers G (or) then $G \subseteq F$
  If all FDs of set G (or) can be inferred from set F, then $G \subseteq F$
- If both $F \subseteq G$ and $G \subseteq F$ are true, then F = G

Note :-

Let A & B are two sets

- $A \subseteq B$ Every element of set A is a member of B {A may or may not be equal to B}
- $A \subset B$ Every element of set A is a member of B and $A \neq B$

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1. If $A \subseteq B$ & $B \subseteq A$, then $A = B$
2. If $A \subseteq B$ & $B \nsubseteq A$, then $A \subset B$
3. If $A \nsubseteq B$ & $B \subseteq A$, then $B \subset A$

#Q. Consider the following FD sets

$F1 = \{A \rightarrow B, B \rightarrow C, AB \rightarrow D\}$ and $F2 = \{A \rightarrow B, B \rightarrow C, A \rightarrow C, A \rightarrow D\}$

Find the relationship between FD sets F1 and F2

$F_1 = A \to B,\ B \to C,\ AB \to D$

$F_2 = A \to B,\ B \to C,\ A \to C,\ A \to D$

① Check if F_1 covers F_2 or not
i.e., $F_2 \subseteq F_1$ or not

FDs of set F_2	Closure w.r.t. F_1	Remark
$A \to B$	$(A)^+_{w.r.t.\ F_1} = \{A, B, C, D\}$	Member of F_1
$B \to C$	$(B)^+_{w.r.t.\ F_1} = \{B, C\}$	Member of F_1
$A \to C$	$(A)^+_{w.r.t.\ F_1} = \{A, B, C, D\}$	Member of F_1
$A \to D$	$(A)^+_{w.r.t.\ F_1} = \{A, B, C, D\}$	Member of F_1

Conclusion: All FDs of F_2 are member of F_1
∴ $F_2 \subseteq F_1$ — Eqⁿ ①

② Check if F_2 covers F_1 or not
i.e., $F_1 \subseteq F_2$ or not

FDs of set F_1	Closure w.r.t. F_2	Remark
$A \to B$	$(A)^+_{w.r.t.\ F_2} = \{A, B, C, D\}$	Member of F_2
$B \to C$	$(B)^+_{w.r.t.\ F_2} = \{B, C\}$	Member of F_2
$AB \to D$	$(AB)^+_{w.r.t.\ F_2} = \{A, B, C, D\}$	Member of F_2

Conclusion: All FDs of F_1 are member of F_2
∴ $F_1 \subseteq F_2$ — Eqⁿ ②

By ① & ② $\boxed{F_1 = F_2}$

#Q. Consider the following FD sets

$F1 = \{A \to B, B \to C, A \to C\}$ and $F2 = \{A \to B, B \to C, A \to D\}$

Find the relationship between FD sets F1 and F2

Check if $A \to C$ is a member of F_2

$(A)^+$ w.r.t $F_2 = \{A, B, C, D\}$

$C \in (A)^+$ w.r.t. F_2

$\therefore A \to C$ is a member of F_2

Hence, $F_1 \subseteq F_2$ — eq^n ①

Check if $A \to D$ is a member of F_1

$(A)^+$ w.r.t. $F_1 = \{A, B, C\}$

$D \notin (A)^+$ w.r.t. F_1

$\therefore A \to D$ is not a member of F_1

Hence $F_2 \not\subseteq F_1$ — eq^n ②

By eq^n ① & ② $F_1 \neq F_2$ but $F_1 \subset F_2$

#Q. Consider the following FD sets

F1 = {A→B, AB→C, D→AC, D→E} and

F2 = {A→BC, D→AE}

Find the relationship between FD sets F1 and F2

$(A)^+ = \{A, B, C\}$

Member of F_2?

$F_1 = \{ A \to B,\ AB \to C,\ D \to AC,\ D \to E \}$

$\therefore F_1 \subseteq F_2$

Member of F_1?

$F_2 = \{ A \to BC,\ D \to AE \}$

$A \to BC = A \to B \,\&\, A \to C$

$D \to AE = D \to A \,\&\, D \to E$

$\therefore F_2 \subseteq F_1$

$F_1 = F_2$



Topic : FD set of a sub-relation

- Let R be a relation with FD set F. We can use the concept of membership test to obtain the FD set of a sub-relation of relation R.

Let $R(A, B, C, D, E)$ is a relation

then $R'(B, C, D)$ can be called a subrelation of R

#Q. Consider a relational schema R(A,B,C,D,E) with FD set

$F = \{A \rightarrow B,\ AB \rightarrow C,\ D \rightarrow AC,\ D \rightarrow E\}$

Find the FD set F1 for sub-relation R1(A,B,E) of R(A,B,C,D,E).

Also find candidate keys for the sub-relation R1(A,B,E).

Soluⁿ on next slide

$R(A\ B\ C\ D\ E)$

$F = \{A \to B,\ AB \to C,\ D \to AC,\ D \to E\}$

↳ Sub-relation of R is $R_1(A, B, E)$

FDs w.r.t. R_1 will specify that what can be determined by the different subsets of attributes from R_1

i.e., $A \to ?$, $B \to ?$, $E \to ?$, $AB \to ?$, $AE \to ?$, $BE \to ?$

$A \to ?$

$(A)^+$ w.r.t. $F = \{A, B, C\}$

i.e., $A \to A\ B\ C$ (A: trivial, C: Not in R_1)

Non-trivial FD w.r.t. $R_1(ABE)$: $A \to B$

$B \to ?$

$(B)^+$ w.r.t. $F = \{B\}$ → Nothing

$E \to ?$

$(E)^+$ w.r.t. $F = \{E\}$ → Nothing

$AB \to ?$

$(AB)^+$ w.r.t. $F = \{A, B, C\}$ → Nothing

$AE \to ?$

$(AE)^+$ w.r.t. $F = \{A, E, B, C\}$ → $AE \to B$

$BE \to ?$

$(BE)^+$ w.r.t. $F = \{B, E\}$ → Nothing

$ABE \to ABE$

will always hold but it is trivial

FD set F_1 w.r.t. sub-relation $R_1(A, B, E)$ is

$F_1 = \{A \to B,\ AE \to B\}$

w.r.t. FD set F_1 Candidate key of $R_1(A, B, E)$ is = AE is C.K

H.W.

#Q. Consider a relational schema R(A,B,C,D,E,F) with FD set

$F = \{AB \rightarrow C, B \rightarrow D, BC \rightarrow A, D \rightarrow EF\}$

Find the FD set F1 for sub-relation R1(A,B,C,D) of R(A,B,C,D,E,F).

Also find candidate keys for the sub-relation R1(A,B,C,D).



2 mins Summary

Topic Identification of candidate keys in a relation

Topic Membership test

Slide

THANK - YOU